

Predictive Modeling of Yes Bank Stock Prices

A concise evaluation of regression architectures for forecasting monthly closing valuations.

Analysis Period: **2005 – 2020**

Observations: **185 Months**

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Project Context & Analytical Goal

CONTEXTUAL GROUNDING & CORE PARAMETERS

- i Historical Volatility:** Yes Bank experienced severe valuation swings post-2018 corporate governance events, creating a highly volatile environment for traditional modeling.
- 🎯 Predictive Objective:** Build a transparent machine learning pipeline to model and predict the final Monthly Closing Price based on historical Open, High, and Low values.
- </> Input Variables:** OLS independent parameters consist of Opening Price, Monthly Peak (High), and Monthly Bottom (Low).



Dataset Profile & Preprocessing

QUANTITATIVE BASELINE & SKEWNESS MITIGATION

Metric Type (INR)	Minimum	Maximum	Average (Mean)	Standard Dev.
Open Price	10.00	375.00	105.54	98.88
High Price	11.25	404.00	116.10	108.83
Low Price	5.55	340.00	94.47	88.51
Close Price (Target)	9.98	367.90	102.63	95.78

Transformation Pipeline: Standard deviation is extremely high relative to the mean, reflecting severe skewness. To stabilize variance and normalize feature distributions, a standard natural log transformation was implemented before fitting the algorithms:

$$X_{\text{transformed}} = \ln (X)$$

Predictive Model Performance

COMPARATIVE ANALYSIS (ADJUSTED R² SCORES)



Interpretation: While standard OLS reports high accuracy, the features exhibit severe multicollinearity (high VIF). **Elastic Net** is chosen as the champion model because it effectively balances L1 (Lasso) and L2 (Ridge) penalties, preventing coefficient inflation and maintaining generalizability.

Analytical Conclusions & Next Steps

SUMMARY OF OPERATIONAL TAKEAWAYS

- ✓ **Pipeline Feasibility:** Preprocessing skewness via natural logarithm transforms proved essential to ensure predictive stability across turbulent timeframes.
- ✓ **Regularization Priority:** Regularized models (specifically Elastic Net) are mathematically required to mitigate the extreme multicollinearity inherent in stock Open/High/Low markers.
- 💡 **Future Parameters:** Transitioning model inputs beyond OHLC by integrating daily sentiment indicators and market volume data will improve predictive accuracy during structural shifts.



Image Sources



https://msftstories.thesourcemediaassets.com/sites/45/2019/09/Yes-bank_765x530.jpg

Source: news.microsoft.com



https://static.vecteezy.com/system/resources/previews/075/595/490/non_2x/dynamic-financial-chart-displays-stock-price-analysis-with-cyan-trend-lines-over-red-candlesticks-forex-graph-reveals-profit-potential-through-blue-overlay-on-volatile-market-vector.jpg

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